01_html_en

Laboratory 01

Holly Grail Layout (3-column layout without Flex/Grid)

A problem from 1998, with over 30 solutions before Flexbox put an end to it. It shows the evolution of web development!

Objective

Classic blog/website layout: Header | (Navigation | Content | Sidebar) | Footer. All columns are the same height regardless of content, without the use of modern Flex, Grid or Table layouts.

Requirements

```
Header: 100% width, fixed height of 80px
Navigation: left column, 200px wide, full page height
Content: centre, flexible width, full height
Aside: right column, 250px wide, full height
Footer: 100% width, 60px height, always at the bottom
```

Responsiveness: At <768px, all columns stack vertically.

Startup template

```html
<div class="page">
  <header>Header: Logo | Menu | Search</header>
  <nav>Nav: Links (20+ Lorem Ipsum items)</nav>
  <main class="content">
    Article about Padel in Poland
    [200+ lorem ipsum words for height testing]
  </main>
  <aside>
    Winter sports
    [15 items for height testing]
```

```
    </aside>
    <footer>© 2026 PadelStats | Poznań</footer>

```

🚫 PROHIBITED techniques:
• display: flex
• display: grid
• display: table
• JavaScript
• Fixed height in %/px (except header/footer)

✅ ALLOWED (classic tricks):
• position: absolute
• float: left/right
• negative margin/padding
• overflow: hidden
• height: 100vh

Proposed solution:

Full HTML

```html
<!DOCTYPE html>
<html lang="pl">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Holly Grail Layout</title>
  <style>
    /* Wklej tutaj CSS wariantu */
  </style>
</head>
<body>
<div class="page">
  <header>🟦 HEADER (80px wys.) | Logo | Menu | Szukaj</header>
```

```html

  <nav>🟨 NAV (200px) - 25 linków dla testu wysokości<br>
    <a href="#">Padel Poznań</a><br>
    <a href="#">Turnieje 2026</a><br>
    <!— +23 lorem linki —>
    <a href="#">Regulamin</a>
  </nav>

  <main class="content">
    <h1>🏓 Historia Padla w Polsce</h1>
    <p>Lorem ipsum dolor sit amet… [200+ słów dla maksymalnego testu wysokości]</p>
    <p>Volkl, Rossignol, Atomic – sprzęt na 2026…</p>
  </main>

  <aside>🟥 ASIDE (250px)<br>
    <h3>Statystyki</h3>
    <ul>
      <li>Mecze: 47</li>
      <!— 15+ pozycji —>
      <li>Średnia: 8.2 pkt</li>
    </ul>
  </aside>

  <footer>⚫ FOOTER - Zawsze na dole © 2026 Politechnika Poznańska</footer>
</div>
</body>
</html>
```

Option 1: FLOAT + NEGATIVE MARGINS (Matthew James Taylor, 2006)

```css
* { margin: 0; padding: 0; box-sizing: border-box; }

body { font-family: system-ui, sans-serif; line-height: 1.6; }

.page {
  overflow: hidden;
```

```
  min-height: 100vh;
  max-width: 1400px;
  margin: 0 auto;
}

/* Float trik */
.content {
  float: left;
  width: 100%;
  margin-left: -200px;
  margin-right: -250px;
  padding: 20px;
  background: white;
}

nav {
  float: left;
  width: 200px;
  margin-left: -100%;
  padding: 20px;
  background: #fef3c7;
  height: 100vh;
}

aside {
  float: right;
  width: 250px;
  margin-right: -100%;
  padding: 20px;
  background: #fee2e2;
  height: 100vh;
}

header, footer {
  clear: both;
  padding: 20px;
  text-align: center;
```

```
  background: #3b82f6;
  color: white;
}

footer { background: #1f2937; }

/* Mobile */
@media (max-width: 768px) {
  .content, nav, aside {
    float: none;
    width: 100%;
    margin: 0;
    height: auto;
  }
}

```

Option 2: ABSOLUTE POSITIONING (Eric Meyer, 1998)

```
* { margin: 0; padding: 0; box-sizing: border-box; }

body {
  font-family: system-ui, sans-serif;
  min-height: 100vh;
}

.page {
  position: relative;
  min-height: 100vh;
  padding: 80px 0 60px;
}

/* Absolute kolumny */
nav {
  position: absolute;
  left: 0;
```

```
  top: 80px;
  bottom: 60px;
  width: 200px;
  padding: 20px;
  overflow-y: auto;
  background: #fef3c7;
}

.content {
  margin: 0 250px 0 200px;
  padding: 20px;
  min-height: calc(100vh - 140px);
  background: white;
}

aside {
  position: absolute;
  right: 0;
  top: 80px;
  bottom: 60px;
  width: 250px;
  padding: 20px;
  overflow-y: auto;
  background: #fee2e2;
}

header {
  position: absolute;
  top: 0;
  left: 0;
  right: 0;
  height: 80px;
  padding: 20px;
  background: #3b82f6;
  color: white;
}

```

```
footer {
  position: absolute;
  bottom: 0;
  left: 0;
  right: 0;
  height: 60px;
  padding: 20px;
  background: #1f2937;
  color: white;
}

/* Mobile */
@media (max-width: 768px) {
  .page { padding: 80px 20px 60px; }
  nav, aside {
    position: static;
    width: 100%;
    height: auto;
  }
  .content { margin: 0; }
}

```

Option 3: PURE CSS (Faux Columns) (Dan Cederholm, 2004)

```
* { margin: 0; padding: 0; box-sizing: border-box; }

body {
  font-family: system-ui, sans-serif;
  min-height: 100vh;
}

.page {
  position: relative;
  min-height: 100vh;
  padding: 80px 0 60px;
```

```
12  }
13
14  /* Absolute kolumny */
15  nav {
16    position: absolute;
17    left: 0;
18    top: 80px;
19    bottom: 60px;
20    width: 200px;
21    padding: 20px;
22    overflow-y: auto;
23    background: #fef3c7;
24  }
25
26  .content {
27    margin: 0 250px 0 200px;
28    padding: 20px;
29    min-height: calc(100vh - 140px);
30    background: white;
31  }
32
33  aside {
34    position: absolute;
35    right: 0;
36    top: 80px;
37    bottom: 60px;
38    width: 250px;
39    padding: 20px;
40    overflow-y: auto;
41    background: #fee2e2;
42  }
43
44  header {
45    position: absolute;
46    top: 0;
47    left: 0;
48    right: 0;
```

```
  height: 80px;
  padding: 20px;
  background: #3b82f6;
  color: white;
}

footer {
  position: absolute;
  bottom: 0;
  left: 0;
  right: 0;
  height: 60px;
  padding: 20px;
  background: #1f2937;
  color: white;
}

/* Mobile */
@media (max-width: 768px) {
  .page { padding: 80px 20px 60px; }
  nav, aside {
    position: static;
    width: 100%;
    height: auto;
  }
  .content { margin: 0; }
}
```